

DeepIso Software (Python)

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1. System requirements

Our software is developed by PyTorch on Windows 10 / Windows 11 with PyCharm (2022.2.3). The package requirements are: numpy = 1.24.3, Pillow = 12.0.0, skimage = 0.0, tifffile = 2023.9.26, torch = 2.6.0+cu126, torchvision = 0.21.0+cu126.

2. Installation guide

- A. The virtual environment and requirements of PyTorch should be installed.
- B. Unzip the “DeepIso_submit.zip” file
- C. Run the Step0~Step3 according to the needs.

3. Pre-preparation of training

- A. Prepare the raw data: the raw data (*.tif files) should be resized to make the real pixel size in the lateral and axial planes the same as each other. For example, the 3D image with pixel of (512, 512, 24) and pixel size of (65, 65, 125) nm should be resized to (512, 512, 46). This step can be easily conducted using ImageJ>Image>Adjust>Size.
- B. Prepare simulated scaled optical transfer function (sOTF): Users can use many open-source PSF generator software to generate the sOTF. If this is hard, using the default simulated sOTF will not compromise the final result normally.
- C. Creating the dataset: Use the “Step0_data_generation.py” file to generate the dataset of 128×128. The “thres” is to judge if the small block contains the effective information. The “raw_data_dir” is the directory to put the raw data. The “data_dir0” is to save the generated dataset, which contains the xoy and xoz blocks.
- D. Correct “sOTF”: Use the “Step1_sOTF_correction.py” file to correct the simulated sOTF. The “Contrast” is to adjust the contrast of the frequency domain. “sOTF_name” is the directory of the simulated sOTF. “XOY_path” and “XOZ_path” are the directories of the dataset that contain the xoy and xoz blocks. The “corrected_sOTF” is the directory to put the corrected sOTF. The intermediate results are put in the “Step1_Degrade” file, which will generate automatically. When the simulated xoz frequency is very similar to the real xoz frequency, the sOTF is corrected successfully, as the Fig .1 shows.

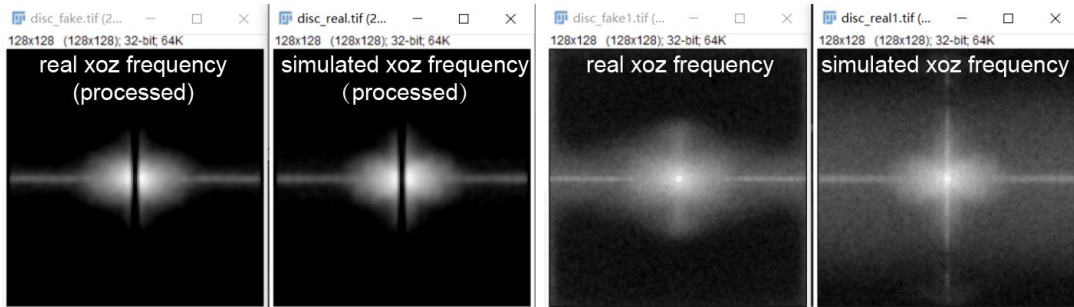


Fig. 1. Intermediate results to generate sOTF

4. Training

Once the sOTF is obtained, the same microscopy system with the same wavelength and pixel size can use this repeatedly, without the need to be corrected for every sample. The training process has two situations:

- A. No/Little axial-down sampling, which means the axial size (before resize) is below 4-fold than the lateral size, and basically satisfies the Nyquist sampling rate. In this situation, you only need to use “Step2_Train_Isotropic_branch.py”. If the noise is overwhelming, “Step2_Train_Isotropic_branch_denoise.py” can replace “Step2_Train_Isotropic_branch.py”. The parameter only needs to put the sOTF directory and XOY/XOZ directories.
- B. Large axial-down sampling, which means the axial size (before resize) is beyond 4 times the lateral size, and 2 to 4 times of Nyquist sampling rate. In this situation, you need to use “Step2_Train_Isotropic_branch.py” and “Step2_Train_Interpolation_branch”. If the noise is overwhelming, “Step2_Train_Isotropic_branch_denoise.py” can replace “Step2_Train_Isotropic_branch.py”. The parameter only needs to put the sOTF directory and XOY/XOZ directories. The “step” in “Step2_Train_Interpolation_branch” means the ratio of axial pixel size and lateral pixel size in the raw data (before resize).

The training process typically takes about 5 minutes, and the model will be saved in the “Step2...” files. The training process with the denoise network will be slightly slow.

5. Predict

The prediction also has two situations:

- A. No/Little axial-down sampling: “Step3_Predict_Isotropic.py” should be used with the model in the previous training. If the denoise model is used, please set “if_denoise” to 1.
- B. Large axial-down sampling: “Step3_Predict_Isotropic_and_interpolation” should be used with the model in the previous training. If the denoise model is used, please set “if_denoise” to 1.

The “input_path” and “output_path” mean the input and output images.

6. Test data

We provide four images to test, containing live_ER, live_Mito, Mouse_actin, and Mouse_actin_down for different situations. The readme.txt contains the guide and information to test.

7. Copyright

We claim an Apache license for DeepIso.